

Erick Watanabe

Long Beach, CA | (562) 673-8014 | erickjwatanabe@gmail.com
erickwatanabe.vercel.app | gitlab.com/SkiSherpa | github.com/SkiSherpa

TECHNICAL SKILLS

Programming Languages | TypeScript, JavaScript ES6+, Python 3, SQL, HTML5, CSS

Front-End | React, Next.js, Vite, Tailwind, Redux Toolkit, DOM manipulation

Back-End | Node.js, Django 4, PostgreSQL, MongoDB, FastAPI

AI/ML & LLMs | RAG, LangChain, Vector databases (Chroma), OpenAI API, LLM API Integration

PROFESSIONAL EXPERIENCE

AI Model Evaluator | Handshake AI 2025 - current

- Evaluate and rank LLM-generated responses across quality dimensions including reasoning, accuracy, and communication, contributing to preference datasets used in model improvement

Staff Scientist | Blue Marble Space Institute of Science 2025

NASA's AI Curation | Next.js, React, TypeScript, Tailwind CSS, FastAPI, Python

- Architected and implemented a full-stack RAG system for scientific document extraction, including FastAPI backend routes with Pydantic validation, LLM-powered retrieval with similarity scoring, and a React frontend with Zustand state management for real-time data display.
- Implemented **accessibility-first** development practices (WCAG compliance, ARIA labels, etc.) ensuring inclusive design for all users across diverse abilities and assistive technologies.
- Led component architecture redesign that transformed monolithic CollapsibleSection components into 12+ reusable, independently maintainable modules, enabling flexible layouts and reducing future feature implementation time by eliminating tight coupling.

NASA's Space Microbial Culture Collection Portal Page | [Link](#) |

Vite, React, TypeScript, Tailwind CSS, Django, Python

- Developed interactive data table using TanStack **React** Table with multi-term search, sorting, and pagination to filter 3,000+ microbial research samples in real-time.
- Architected solution to serve React SPA through Django by implementing custom view logic for static assets and data files, configuring Vite build process with server-specific base paths, and establishing catch-all URL patterns to support client-side routing while maintaining API endpoint access.

Full Stack Engineer Contractor | Valter Tennis | [Link](#) | Next.js, Firebase, TypeScript 2024

- Enhanced search engine optimization (SEO) by applying correct tags, utilizing keywords, a sitemap, and robot's file, resulting in an 85% increase in traffic originating from Google searches.
- Created a responsive main navigation bar for the website, ensuring seamless functionality across devices, with 54% of clients accessing the site through a mobile browser.

Visiting Assistant Professor of Math & Physics | Loyola Marymount University 2017 - 2022

EDUCATION

Hack Reactor | Advanced Software Engineering Certificate 2023

California State University, Long Beach | M.A. Experimental Physics 2016

The University of California, Merced | B.A. Physics 2011